

Cyber Sentinel – Keylogger Detection System

Project Name: Cyber Sentinel (CS_KEY_LOGGER)

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1. Abstract

Cyber Sentinel is a specialized security tool designed to detect and monitor unauthorized keylogging activities on a computer system. It provides real-time detection of suspicious processes and alerts users to potential threats, enhancing system-level security and data privacy.

2. Problem Statement

Keyloggers are a form of spyware capable of silently capturing user keystrokes. Traditional antivirus solutions may fail to detect custom or lightweight keylogging scripts. This project addresses the need for a dedicated, behavior-focused detection system.

3. Objectives

- Real-time detection of suspicious keylogging activity.
- Immediate user alerts upon threat detection.
- Identification and termination of malicious processes.
- Persistent logging for forensic analysis.
- A user-friendly, web-based monitoring interface.

4. System Architecture

4.1 Backend (Python)

The backend leverages Python with the psutil library to scan system processes. Flask is used to expose API endpoints for scanning, logging, and process management. Detection logs are stored persistently in JSON format.

4.2 Frontend (HTML/CSS/JavaScript)

The frontend consists of a responsive web dashboard designed with modern UI principles. It provides real-time status updates, threat visualization, and user interaction without page reloads.

5. Features

- Live scanning of active system processes.
- Interactive dashboard with real-time system status.

- Persistent detection and activity logs.
- Backend-supported process termination.
- Visual feedback using status indicators and alerts.

6. Technology Stack

- Python 3.x
- Flask Web Framework
- psutil System Library
- HTML5, CSS3, JavaScript
- JSON for data storage

7. Conclusion

Cyber Sentinel demonstrates an effective host-based intrusion detection approach focused on keyloggers. By combining a robust backend detection engine with an intuitive web interface, the system makes security monitoring accessible and actionable. Future enhancements may include heuristic and network-based analysis.